

32) A says "I am the Knight", B says ~~"A is telling the truth"~~
B says "A is not the Knave," and C says "B is not the Knave".

Soln: Let A be the Knight. $\therefore$ "I am the Knight" is true which is consistent with the fact that a Knight always speaks the truth.
B says "A is not the Knave" which is True. $\therefore$ B must be the spy.
 $\therefore$ C is the Knave. C says "B is not the Knave" which is True. This contradicts the fact that a Knave always lies.

$\therefore$ A cannot be the Knight

Let A be the spy. $\therefore$ "I am the Knight" is False which is consistent with the fact that a spy can lie.
B says "A is not the Knave" which is True. $\therefore$ B must be the Knight.
 $\therefore$ C is the Knave. C says "B is not the Knave" which is True. This contradicts the fact that a Knave always lies.

$\therefore$ A cannot be the spy

So, A must be the Knave. $\therefore$ "I am the Knight" is False which is consistent with the fact that a Knave always lies.

B says "A is not the Knave" which is False. $\therefore$ B must be a spy.

C says "B is not the Knave" which is True. $\therefore$ C must be a Knight.

Unique solution: $A \rightarrow \text{Knave}$ $B \rightarrow \text{spy}$ $C \rightarrow \text{Knight}$

33) A says "I am the Knight", B says "I am the Knight", and C says "I am the Knight."

Soln: A Knight, who always speaks the truth can say "I am the Knight" because it is True. A spy, who can lie sometimes can say "I am the Knight" because it is False. A Knave, who ~~can~~ always lie can say "I am the Knight" because it is False. The following are the possible solutions:

Soln 1: $A \rightarrow \text{Knight}$, $B \rightarrow \text{spy}$, $C \rightarrow \text{Knave}$

Soln 2: $A \rightarrow \text{Knight}$, $B \rightarrow \text{Knave}$, $C \rightarrow \text{spy}$